

Lab Report

Name: Samura Huda

Id: 22201205

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Faculty Name: Shabnam Ahsan (SHBM)

Introduction:

The ABO blood grouping system was discovered by Austrian scientist Karl Landsteiner in 1900. He was the first person who observed that mixing blood from different individuals sometimes caused clumping, or agglutination. Later he identified the A, B, and O blood types which brought a revolution in the field of medical science.

According to Kind (1962) based on the presence of different antigens in red blood cell ABO blood grouping system categorized human blood into 4 major groups. Which are A, B, AB and This classification is determined through blood typing tests, where blood samples are mixed with antibodies specific to A and B antigens. Depending on the reaction (clumping or agglutination), the blood type is determined.

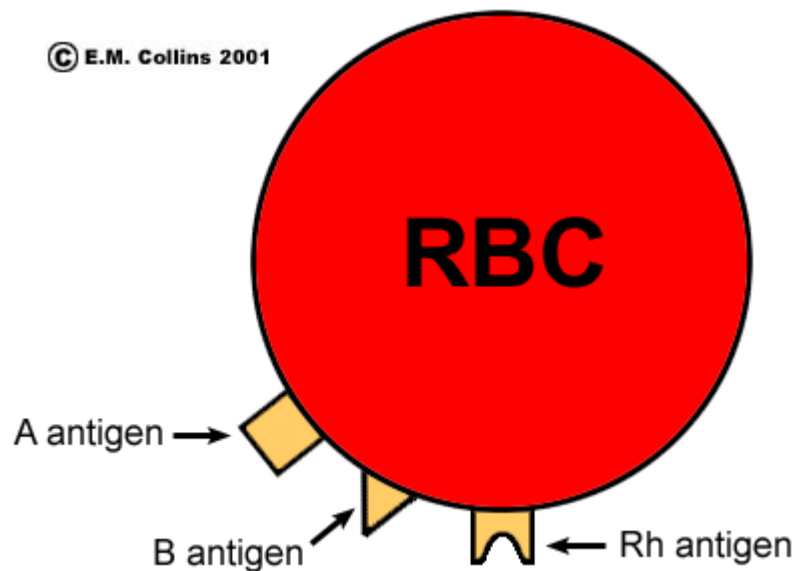


Figure: Red blood cell

Blood type A: it has A antigens. Blood type A has anti-B antibodies.

Blood type B: it has B antigens. Blood type B has anti-A antibodies.

Blood type AB: it has both A and B antigens.

Blood type O: it has neither A nor B antigens. and blood type O has both anti-A and anti-B antibodies.

According to Offner, S (2015) Before discovering ABO blood grouping in the early days of surgery when some surgical patients mysteriously died after receiving a transfusion of what we now know was blood of the wrong type. It is significant because it determines blood compatibility for transfusions. Understanding ABO blood types helps prevent deathly reaction in body when wrong group's blood is transfused. Mismatched blood types can lead to organ rejection and failure. By matching the ABO blood types of donors and recipients, transplant success rates increase significantly.

Blood group AB is the universal recipient and can receive blood from all other blood groups (A, B, AB, and O).

Blood group O is the universal donor and can donate blood to all other blood groups (A, B, AB, and O).

The Rh system, also known as the Rhesus system, is a blood group system that categorizes blood based on the presence or absence of the Rh antigen (or D antigen) on the surface of red blood cells. Those who possess the Rh antigen are classified as Rh-positive, while those lacking it are considered Rh-negative. The importance of the Rh system lies in its significance for blood transfusions and pregnancy. Rh-positive individuals can receive both Rh-positive and Rh-negative blood, while Rh-negative individuals should only receive Rh-negative blood to avoid the development of Rh antibodies, which could lead to severe immune reactions upon subsequent exposures to Rh-positive blood. So, in blood transfusion it is very significant.

LAB: BLOOD TYPING (ABO SYSTEM)

List of materials:

- Microscopic slides
- Blood lancets
- Blood samples
- Blood grouping test kits
- Sterile toothpicks

Procedure:

- Divided a microscopic slide in three parts using a pen. Marked one part as A, second part as B and last part as D.
- Wiped one finger with alcohol saturated cotton and pricked with a sterile needle to draw blood.
- Put three drops of blood on the three separate parts of the slide.
- Put a drop of anti-A serum on part A, a drop of anti-B on part B and anti-D on part D.
- Mixed the serum and blood with toothpicks
- Then waited for the changes in A, B and D part

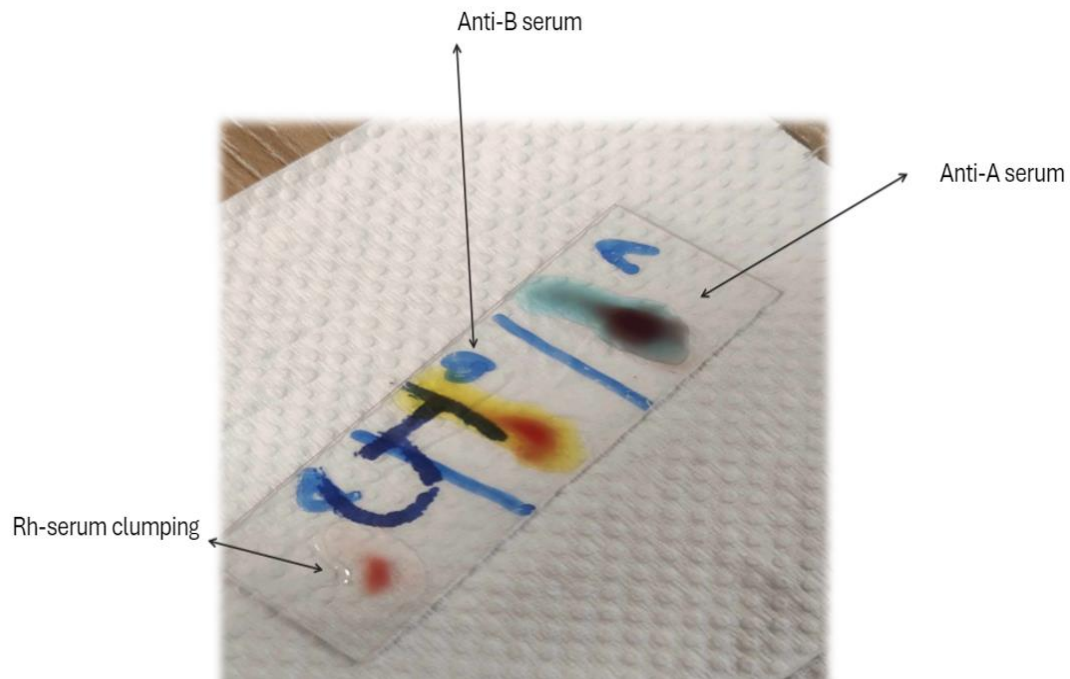


figure: Blood grouping system

Observation and Result:

Here in the slide we saw in part A that there is no blood clumping with using anti A serum. Also the result is same for part B. After using anti B serum there is no blood clumping. But in the section D, using Rh serum we saw that there was a little bit blood clumping which is for Rh antigens on red blood cell. Though it was not agglutinated but the clumping was clear. In the blood grouping table, we also see that the result for D serum is similar to mine. Here blood is also clumping.

HOW TO READ YOUR RESULTS

BLOOD TYPE	ANTI-A	ANTI-B	ANTI-D	CONTROL
O-POSITIVE				
O-NEGATIVE				
A-POSITIVE				
A-NEGATIVE				
B-POSITIVE				
B-NEGATIVE				
AB-POSITIVE				
AB-NEGATIVE				
INVALID				

Figure: Blood grouping table

Discussion:

According to the observation, the blood sample lacks both A and B antigens on the red blood cells, indicating blood type O. However, there is clumping observed in part D of the slide, indicating the presence of the Rh antigen, resulting in a positive Rh factor. In the ABO system, individuals produce antibodies against antigens A or B, except for those with blood type O, who lack these antigens and do not produce corresponding antibodies. Therefore, individuals with O positive blood type, being universal donors, can safely donate blood to individuals with any other blood type (A, B, AB) without causing adverse reactions. However, individuals with O blood type produce antibodies against both A and B antigens, making them unable to receive blood from individuals with A, B, or AB blood types. Instead, O positive individuals can receive blood from compatible donors, including other O positive individuals or O negative individuals. In my experiment I clearly saw the clumping, but it was not Agglutinating. The reason might be that the amount of sample was very little also we had to wait more to see the exact outcome.

Reference

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